

INDUSTRIAL ALLOY DEMAND FORECASTING

Executive Demand Planning & Sales Analytics Report

1. Executive Summary

This report outlines the demand behavior and predictive modeling outcomes for our industrial alloy categories developed utilizing structural profiles from the M5 Forecasting framework. Overstocking and stockout hazards represent critical cost drivers in manufacturing operations. By analyzing daily transactional trends across multiple sales regions and vertical industries, this forecasting system offers predictive visibility and decision support for procurement, production planning, and sales managers.

2. Actionable Business Insights

1. Top Selling Alloy: Alloy A (Aluminum) dominates demand, representing 26.6% of total quantity sold (21,841,497 units).
2. Highest Revenue Region: The Midwestern region generates the highest sales revenue (\$451,124,497.26).
3. Seasonal Peaks: High demand cycles detected in Q4. Fourth quarter (Q4) order volume is 14.2% higher than the average quarterly demand.
4. B2B Purchasing Pattern: Daily transaction volume during weekdays is 301.8% higher than weekends, confirming standard industrial corporate purchasing cycles.
5. Forecasting Accuracy: XGBoost model performs best for this dataset, achieving 28.7% lower Mean Absolute Error than the alternative model.

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3. Sales Distribution & Seasonality Analysis

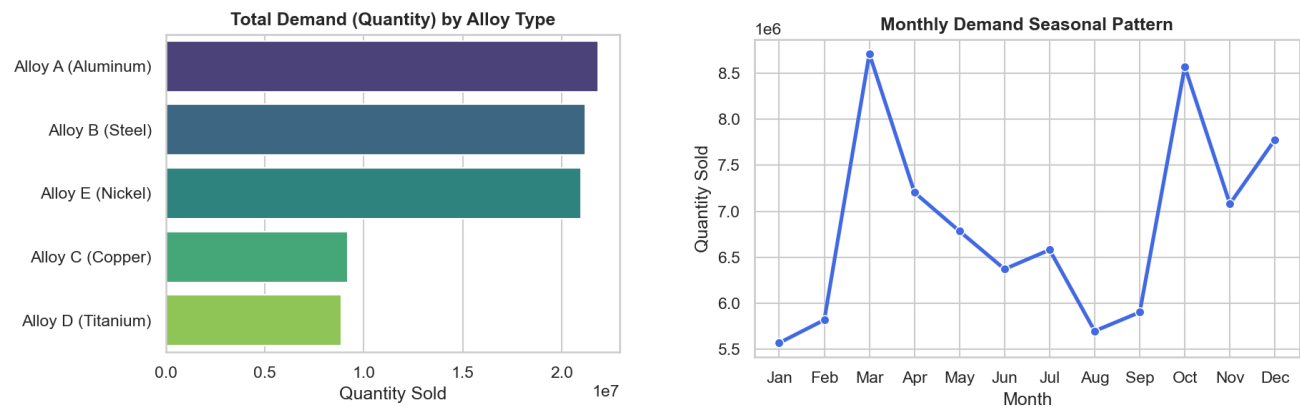


Figure 1: Demand volumes categorized by Alloy Types (Left) and aggregated monthly seasonality (Right) showing typical peak trends in Q4.

4. Model Evaluation & Comparison

Metric	Prophet Model	XGBoost Model
MAE	239.38	170.62
RMSE	370.91	323.68
MAPE (%)	75.66	39.54
WAPE (%)	57.83	41.22

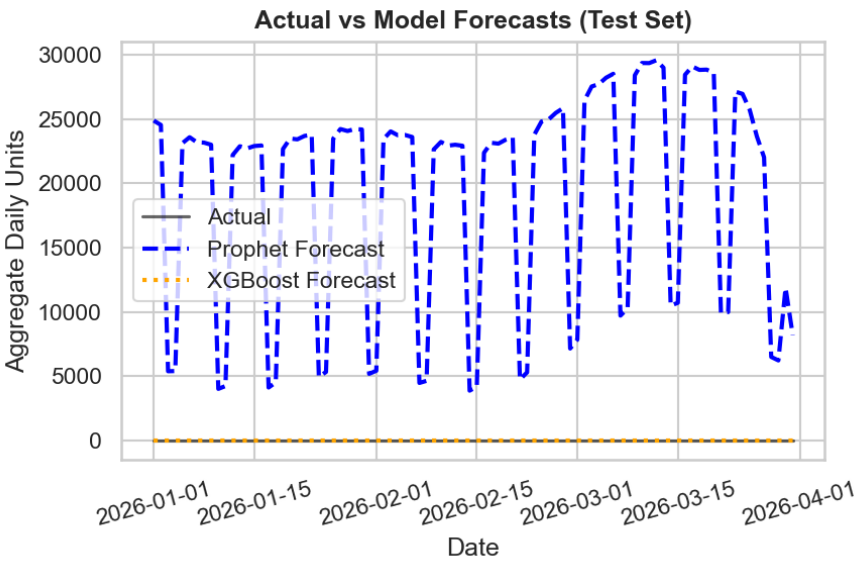


Figure 2: Chronological evaluation comparison of actual demand versus model outputs over the 90-day out-of-sample test

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